

Medical Information

Reference Guide

For Testing & Educational Purposes

This document is for informational purposes only and does not constitute medical advice. Always consult a qualified healthcare professional.

1. Normal Vital Signs — Adult Reference

Parameter	Normal Range	Clinical Notes
Heart Rate (Pulse)	60 – 100 bpm	< 60 bpm (bradycardia) or > 100 bpm (tachycardia)
Blood Pressure	< 120/80 mmHg	Stage 1 HTN: 130–139/80–89 mmHg
Respiratory Rate	12 – 20 breaths/min	< 12 or > 20 breaths/min
Body Temperature	36.1 – 37.2 °C	Fever: > 38.0 °C Hypothermia: < 35.0 °C
Oxygen Saturation (SpO2)	95 – 100 %	< 90 % requires urgent assessment
Blood Glucose (fasting)	70 – 99 mg/dL	Diabetes: ≥ 126 mg/dL on two tests

Note: Values may vary slightly by age, sex, fitness level, and laboratory reference ranges.

2. Common Medical Conditions — Overview

Hypertension (High Blood Pressure)

A chronic condition where the force of blood against artery walls is consistently too high. Often asymptomatic; long-term complications include heart disease, stroke, and kidney damage.

Common Symptoms:

- Persistent headache
- Dizziness or lightheadedness
- Blurred vision
- Shortness of breath
- Chest pain (severe cases)

Management: Lifestyle changes (diet, exercise, salt reduction), antihypertensive medications (ACE inhibitors, beta-blockers, calcium channel blockers, diuretics).

Type 2 Diabetes Mellitus

A metabolic disorder characterised by high blood sugar due to insulin resistance or insufficient insulin production by the pancreas.

Common Symptoms:

- Increased thirst and urination
- Fatigue and blurred vision
- Slow-healing wounds
- Frequent infections
- Tingling in hands/feet

Management: Blood sugar monitoring, dietary modification, weight management, metformin or other hypoglycaemic agents, insulin therapy in advanced cases.

Asthma

A chronic respiratory condition involving inflammation and narrowing of the airways, leading to recurring episodes of wheezing, breathlessness, and coughing.

Common Symptoms:

- Wheezing
- Shortness of breath
- Chest tightness
- Coughing (worse at night or early morning)

Management: Inhaled corticosteroids (preventer), short-acting beta-agonists (reliever/rescue), avoidance of triggers (allergens, smoke, cold air).

3. Commonly Used Medications

Medication	Indication	Typical Adult Dose	Key Precautions
Paracetamol (Acetaminophen)	Pain relief, fever reduction	500–1000 mg every 4–6 h (max 4 g/day)	Hepatotoxicity in overdose
Ibuprofen	Anti-inflammatory, pain, fever	200–400 mg every 6–8 h with food	GI ulcers, renal impairment, avoid in heart failure
Amoxicillin	Bacterial infections	250–500 mg every 8 h for 5–10 days	Allergy (rash, anaphylaxis), GI upset
Metformin	Type 2 Diabetes	500 mg twice daily (titrate to 2 g/day)	GI upset, lactic acidosis (rare), hold before contrast
Lisinopril	Hypertension, heart failure	5–40 mg once daily	Dry cough, hyperkalaemia, angioedema
Atorvastatin	High cholesterol	10–80 mg once daily (evening)	Myopathy, elevated liver enzymes
Salbutamol (Albuterol)	Asthma / bronchospasm rescue	1–2 puffs (100 mcg each) as needed	Tachycardia, tremor, hypokalaemia (excess use)
Omeprazole	Acid reflux, peptic ulcer	20–40 mg once daily before meals	Hypomagnesaemia (long-term), C. diff risk

■ Always follow prescriber instructions. Never adjust or stop medication without medical advice. Check for drug interactions and contraindications before administering.

4. Emergency First Aid — Quick Reference

Cardiac Arrest — CPR Steps

1. Call emergency services (911 / 112 / 15) immediately.
2. Place heel of hand on centre of chest; interlock fingers.
3. Push down hard and fast: 100–120 compressions/min, 5–6 cm depth.
4. Allow full chest recoil between compressions.
5. Give 2 rescue breaths after every 30 compressions (if trained).
6. Continue until AED arrives, help arrives, or patient recovers.

Choking — Heimlich Manoeuvre

1. Ask 'Are you choking?' — if they cannot speak/cough, act.
2. Stand behind the person, lean them slightly forward.
3. Give 5 firm back blows between shoulder blades.
4. Place fist above navel; give 5 upward abdominal thrusts.
5. Alternate back blows and abdominal thrusts until object dislodges.
6. If unconscious, begin CPR and look for object before each breath.

Severe Bleeding

- 1. Apply firm, direct pressure with a clean cloth or bandage.
- 2. Do not remove the cloth — add more on top if soaked.
- 3. Elevate the injured limb above heart level if possible.
- 4. Apply a tourniquet 5–7 cm above wound for life-threatening limb bleed.
- 5. Note the time tourniquet was applied; do not remove it.
- 6. Call emergency services immediately.

5. Body Mass Index (BMI) Classification

BMI is calculated as: $\text{weight (kg)} \div \text{height}^2 \text{ (m)}$. It is a screening tool and does not directly measure body fat or health.

BMI (kg/m²)	Category	Recommended Action
< 18.5	Underweight	Nutritional assessment, possible malnutrition risk
18.5 – 24.9	Normal weight	Maintain healthy lifestyle
25.0 – 29.9	Overweight	Diet and physical activity advised
30.0 – 34.9	Obesity Class I	Medical evaluation; cardiovascular risk elevated
35.0 – 39.9	Obesity Class II	High risk; specialist referral often indicated
≥ 40.0	Obesity Class III	Very high risk; bariatric evaluation may be needed

BMI has limitations: it does not account for muscle mass, bone density, age, sex, or ethnicity. Waist circumference and other assessments should complement BMI for a complete picture.